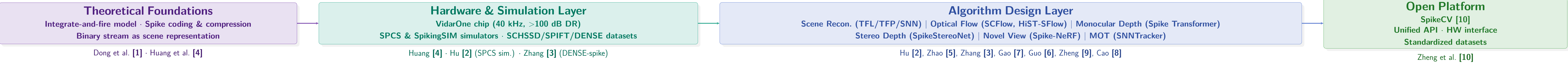
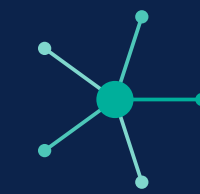


# Spike Cameras for High-Speed Computer Vision

A Survey of 10 Works (2017-2026)

Dong [1] · Huang [4] · Hu [2] · Zhao [5] · Zhang [3] · Gao [7] · Guo [6] · Zheng [9,10] · Cao [8]



## Sensor Principles Integrate-and-Fire Model [1][4]

Spike cameras mimic the **retinal fovea**: each pixel accumulates photons and fires a binary spike when  $A(x, t) \geq \varphi$ , then resets. Polling at  $\Delta t = 25 \mu s$  yields **40 000 Hz** binary streams.

Dong et al. 2017<sup>[1]</sup> TFL & TFP:

- **TFL (Texture from Latencies)**: Reconstruct brightness inversely proportional to the time interval between successive spikes. Fails in extremely bright regions if spikes are faster than hardware clock.
- **TFP (Texture from Playback)**: Counts spikes inside a temporal window. Fails in dark regions unless windows are large, leading to motion blur.

→ Early foundations for modeling the light intensity formulation.

**VidarOne chip**<sup>[4]</sup>:

- **400×250** resolution, **>100 dB** dynamic range, **370 mW** power.
- Output rate: **40 kHz**. Unlike conventional cameras capturing average radiance continuously, each pixel in VidarOne makes asynchronous threshold-based decisions.

## Sensor Comparison: Frame / Event / Spike Cameras

| Property      | Frame     | Event      | Spike         |
|---------------|-----------|------------|---------------|
| Temporal res. | 30240 fps | ~1 MHz     | <b>40 kHz</b> |
| Motion blur   | Severe    | None       | None          |
| Scene texture | Full      | Edges only | Full          |
| HDR           | Poor      | Good       | >100 dB       |
| Pixel Output  | ADC (Int) | Binary     | Binary stream |
| Redundancy    | High      | Low        | High          |

## Key Evaluation Datasets

To train and benchmark spike vision models, several synthetic and real datasets have been introduced:

- **SPIFT & PHM**<sup>[2][5]</sup>: SPIFT (*SpIkingly Flying Things*) features objects flying in random backgrounds (like FlyingChairs) for robust optical flow training. **PHM** (*Photo-realistic Highspeed Motion*) offers well-designed scenes with high-speed camera motions and varying illuminations.
- **DENSE-spike**<sup>[3]</sup>: A synthetic weather-condition dataset containing RGB imagery, dense spike streams, and accurate ground-truth depth maps for monocular depth estimation (MDE).
- **Outdoor-spike**<sup>[3]</sup>: Real-world sequences captured on traffic roads and city streets directly via a hardware spiking camera to validate real-world scene generalization.

## SpikeCV: Open Spike Vision Platform<sup>[10]</sup>

Zheng et al., 2026<sup>[10]</sup> first open-source CV platform for spike cameras.

- SpikeStream: unified API for offline & live hardware
- Standardized datasets: flow, depth, detection, tracking
- Hardware: SpikeSEE cameras (megapixel color, 3rd gen)

## Dense Prediction I Optical Flow [2][5]

**Challenge**: Spike density fluctuates (Poisson photon arrival) · noisy feature matching.

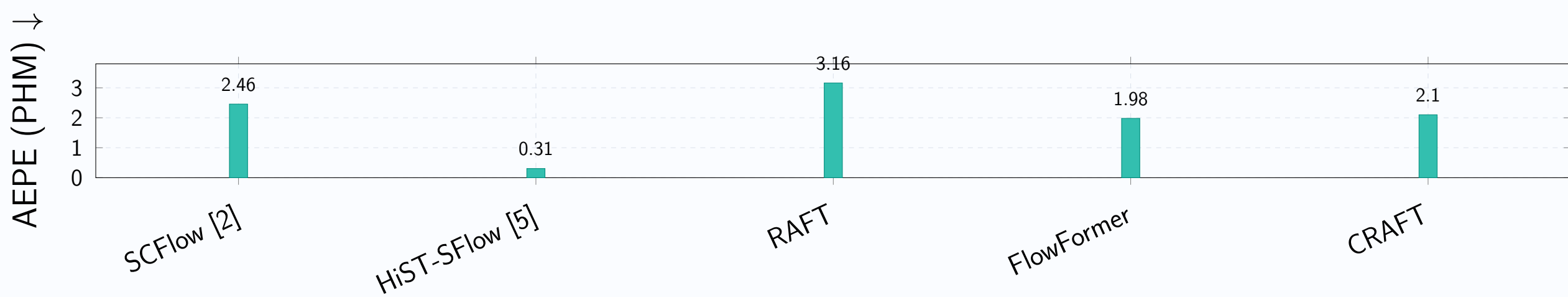
**SCFlow**<sup>[2]</sup> (Hu, CVPR 2022)

FATW: adaptive temporal window.

- 0.80 M params (RAFT: 5.40 M)

- SPIFT AEE: **2.457**

**Insight**: Conventional feature matching fails under random, fluctuating spike densities (Poisson photon arrival). HiST-SFlow resolves this by constructing dense representations across multiple spatial-temporal scales prior to cost volume calculation.



## Dense Prediction II Depth Estimation [3][7]

**Spike Transformer**<sup>[3]</sup>

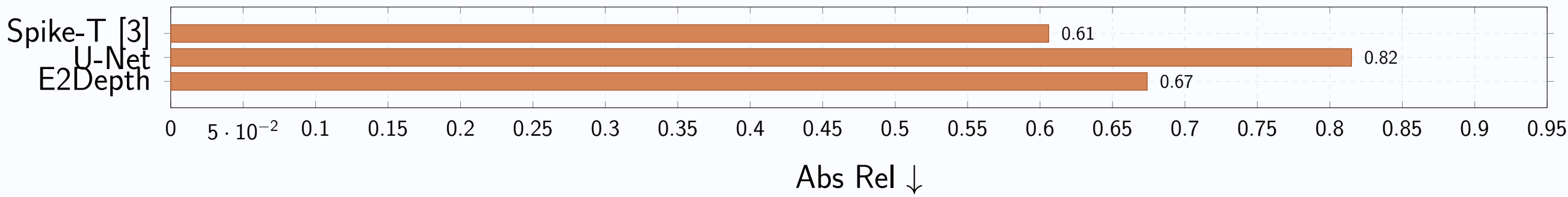
(Zhang, ECCV 2022)

First monocular depth from spikes.

- DENSE-spike Abs Rel: **0.606**

- $\delta < 1.25$ : **0.682**

**Insight**: Spike Transformer provides the foundational architecture to effectively mine structural/spatial features directly from continuous, irregular binary streams without reconstructing images first. SpikeStereoNet extends this heavily into the stereo domain.



**HiST-SFlow**<sup>[5]</sup> (Zhao, AAAI 2024)

Hierarchical spatial-temporal fusion.

- 2030% error reduction

- PHM AEPE: **0.305**

**SpikeStereoNet**<sup>[7]</sup>

(Gao, 2026)

First stereo depth from spikes.

- RAFT-Stereo + Recurrent SNN

- Robust with 10-50% data

## Cross-Cutting Adaptive Stream Processing<sup>[8]</sup>

**SpikeSlicer**<sup>[8]</sup> (Cao, NeurIPS 2024): SNN as adaptive event stream slicer. Fixed-length slicing causes information loss or redundancy; *SPA-Loss* trains SNN to fire at optimal slice points.

- **SPA-Loss Modulation**: Guides the SNN to spike and segment sequences effectively by regulating neuron membrane potentials based on semantic density.
- **Feedback-Update Training**: Refines SNN slicing decisions dynamically using downstream Artificial Neural Network (ANN) accuracy loss feedback.
- **Performance**: Tracking **+11.9%**, Recognition **+19.2%**, overhead **+0.32%**. Creates a novel low-energy SNN-ANN cooperative processing paradigm.

## High-Speed Vision System<sup>[4]</sup>

Huang, 2023<sup>[4]</sup> end-to-end system on VidarOne chip.

**Pipeline**:

- Scene reconstruction (TFW/TFI)
- Detection (LIF neuron, STP filtering)
- Tracking (CANN trajectory prediction)
- Processing: **20 811 Hz**

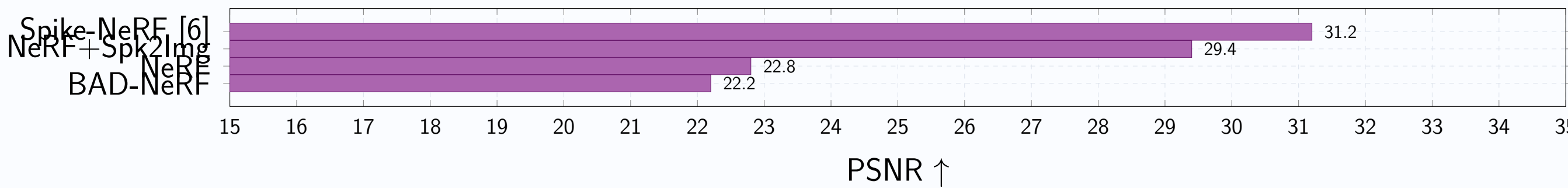
**Performance**:

|             |               |
|-------------|---------------|
| MOTA        | <b>96.32%</b> |
| ID Switches | <b>0</b>      |
| Power       | 2.254 W       |
| Max speed   | 40 m/s        |

## Neural Scene Representation Spike-NeRF<sup>[6]</sup>

Guo, ICME 2024<sup>[6]</sup> first NeRF from spike streams. Standard NeRF fails on high-speed scenes (motion blur); spike cameras deliver sharp 40 kHz data.

**Innovations**: Spiking volume renderer, spike loss, spike masks. By deriving volumetric fields directly from the uncompressed 40 kHz binary stream, Spike-NeRF successfully bypasses the severe error accumulation common in conventional spike-to-image reconstruction chains.



## High-Level Vision Multi-Object Tracking<sup>[9]</sup>

**SNNTracker**<sup>[9]</sup> (Zheng, TPAMI 2026) first fully-SNN MOT for spike cameras.

**Architecture Components**:

- **Adaptation & Motion Estimation**: Encodes spatio-temporal dynamics directly without frames.
- **Attention-based Saccade**: Uses Dynamic Neural Fields (DNF) for low-latency target detection.
- **WTA + STDP Tracking**: Real-time online learning via Winner-Takes-All mechanisms, achieving state-of-the-art accuracy with *zero labeled data*.

**Impact**: Overcomes traditional tracker failures on fast target rotations or abrupt occlusions by continuously updating trajectory confidence maps inside an SNN framework.

## Open Problems and Future Directions

- **Real-world 3D benchmarks**: Spike-NeRF<sup>[6]</sup> and SpikeStereoNet<sup>[7]</sup> need real datasets with accurate depth/pose.
- **End-to-end SNN stacks**: No paper yet combines adaptive slicing, dense prediction, and tracking in one SNN.
- **Megapixel & color**: SpikeCV<sup>[10]</sup> documents color megapixel hardware, but all algorithms assume grayscale low-res input.
- **SNN-ANN Synergy Paradigms**: Leveraging SNNs strictly as high-speed front-end processors (e.g. SpikeSlicer) for powerful downstream ANNs provides an extremely viable energy-efficient workflow.



## Appendix 1: Full References

- [1] Dong, Siwei, Tiejun Huang, and Yonghong Tian. Spike Camera and Its Coding Methods. *2017 Data Compression Conference (DCC)*, April 2017, 437437. <https://doi.org/10.1109/DCC.2017.69>.
- [2] Hu, Liwen, Rui Zhao, Ziluo Ding, et al. Optical Flow Estimation for Spiking Camera. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, 1782332. <https://doi.org/10.1109/CVPR52688.2022.01732>.
- [3] Zhang, Jiyuan, Lulu Tang, Zhaofei Yu, Jiwen Lu, and Tiejun Huang. Spike Transformer: Monocular Depth Estimation for Spiking Camera. In *Computer Vision ECCV 2022*, vol. 13667, edited by Shai Avidan, Gabriel Brostow, Moustapha Cissé, Giovanni Maria Farinella, and Tal Hassner. Lecture Notes in Computer Science. Springer Nature Switzerland, 2022. [https://doi.org/10.1007/978-3-031-20071-7\\_3](https://doi.org/10.1007/978-3-031-20071-7_3).
- [4] Huang Tiejun, Zheng Yajing, Yu Zhaofei, et al. 1000€ Faster Camera and Machine Vision with Ordinary Devices. *Engineering* 25 (June 2023): 11019. <https://doi.org/10.1016/j.eng.2022.01.012>.
- [5] Zhao, Rui, Ruiqin Xiong, Jian Zhang, Xinfeng Zhang, Zhaofei Yu, and Tiejun Huang. Optical Flow for Spike Camera with Hierarchical Spatial-Temporal Spike Fusion. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, no. 7 (2024): 7496504. <https://doi.org/10.1609/aaai.v38i7.28581>.
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- [8] Cao, Jiahang, Mingyuan Sun, Ziqing Wang, et al. Spiking Neural Network as Adaptive Event Stream Slicer. arXiv:2410.02249. Preprint, arXiv, March 24, 2025. <https://doi.org/10.48550/arXiv.2410.02249>.
- [9] Zheng, Yajing, Chengen Li, Jiyuan Zhang, Zhaofei Yu, and Tiejun Huang. SNNTracker: Online High-Speed Multi-Object Tracking With Spike Camera. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48, no. 1 (2026): 62438. <https://doi.org/10.1109/TPAMI.2025.3610696>.
- [10] Zheng, Yajing, Jiyuan Zhang, Rui Zhao, et al. SpikeCV: Open a Continuous Computer Vision Era. *Science China Information Sciences* 69, no. 3 (2026): 132106. <https://doi.org/10.1007/s11432-023-4565-6>.



## Appendix 2: AI Usage

- ▶ Tried (overall thought → DeepSeek V4 → detailed poster description → GPT Image 2 → poster). (It was not available to upload 10 files to GPT Image 2 without Pro membership.)
- ▶ But it was vague and hard to adjust.
- ▶ Tried (overall thought → Claude Code → poster tex).
- ▶ It worked well despite some annoying display bug (esp. asymmetrical margin, undersized fonts).
- ▶ Fixed and refined with Gemini 3.1 Pro from Copilot.
- ▶ Updated reference items manually.
- ▶ Finally checked (given 10 papers and full tex file) by DeepSeek V4.